Comparative Study of Label Encoding and One-Hot Encoding Techniques for Categorical Data

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Dataset Description and objective

```python
data = {
    'Gender': ['Male', 'Female', 'Female', 'Male', 'Female'],
    'AppointmentType': ['General', 'Emergency', 'Consultation', 'General', 'Emergency'],
    'City': ['Kottayam', 'Kochi', 'Calicut', 'Kochi', 'Kottayam']
}

df = pd.DataFrame(data)

print("Original Dataset:")
display(df)
```

Original Dataset:

	Gender	AppointmentType	City
0	Male	General	Kottayam
1	Female	Emergency	Kochi
2	Female	Consultation	Calicut
3	Male	General	Kochi
4	Female	Emergency	Kottayam

The objective of this task is to understand and compare Label Encoding and One-Hot Encoding techniques for converting categorical data into numerical form. The task focuses on analysing how each method transforms the dataset and identifying appropriate use cases for each encoding technique.

Datasets often contain categorical features such as Gender, AppointmentType, and City, which cannot be directly used in machine learning models. Encoding techniques are required to convert these categories into numerical values. Different encoding methods impact the dataset structure and interpretation differently. This task is designed to provide practical understanding of Label Encoding and One-Hot Encoding and their effects on data pre-processing.

Label Encoding

It is an encoding technique that is used to convert categorical data into numerical data. It gives a unique integer value to each category. The original size of the data does not change in label encoding. It is suitable for ordinal category. It use low memory and also have lower computational cost.

```
#label encoding
le = LabelEncoder()
```

```
label_df['Gender'] = le.fit_transform(label_df['Gender'])
label_df['AppointmentType'] = le.fit_transform(label_df['AppointmentType'])
label_df['City'] = le.fit_transform(label_df['City'])

print("Label Encoded Dataset:")
display(label_df)
```

Label Encoded Dataset:

	Gender	AppointmentType	City
0	1	2	2
1	0	1	1
2	0	0	0
3	1	2	1
4	0	1	2

One-Hot Encoding

In one hot encoding a it creates separate binary columns for each category, where 1 represents the presence of a category and 0 represents its absence. It is preferred for nominal categorical data. This can lead to higher number of columns . There is also high usage of memory and higher computation cost.

```
#One-Hot Encoding
onehot_df = pd.get_dummies(df)
```

```
print("One-Hot Encoded Dataset:")
display(onehot_df)
```

One-Hot Encoded Dataset:

	Gender_Female	Gender_Male	AppointmentType_Consultation	AppointmentType_Emergency	AppointmentType_General	City_Calicut	City_Kochi	City_Kottayam
0	False	True	False	False	True	False	False	True
1	True	False	False	True	False	False	True	False
2	True	False	True	False	False	True	False	False
3	False	True	False	False	True	False	True	False
4	True	False	False	True	False	False	False	True

Comparative Analysis

```
#comparative analysis
print("Original Dataset Shape:", df.shape)
print("Label Encoded Shape:", label_df.shape)
print("One-Hot Encoded Shape:", onehot_df.shape)
```

```
Original Dataset Shape: (5, 3)
Label Encoded Shape: (5, 3)
One-Hot Encoded Shape: (5, 8)
```

```
comparison = pd.DataFrame({
    'Dataset': ['Original', 'Label Encoded', 'One-Hot Encoded'],
    'Rows': [df.shape[0], label_df.shape[0], onehot_df.shape[0]],
    'Columns': [df.shape[1], label_df.shape[1], onehot_df.shape[1]]
})
display(comparison)
```

	Dataset	Rows	Columns
0	Original	5	3
1	Label Encoded	5	3
2	One-Hot Encoded	5	8

The shape of the dataset is retained in label encoding while the dimension increase in one hot encoding. The memory usage is low and computational power is low for label encoding while both are high for one hot encoding. Label encoding is used for ordinal dataset while one hot encoding is used for ordinal dataset. An artificial ordering is created when using label encoding while one hot encoding does not do that.

Conclusion

Label encoding maintains the original dataset size but it may create misleading ordinal relationship between categories. While one hot encoding avoid this issue it increases the size of the dataset . The nature of the categorical dataset decides which encoding method to use. For the given dataset the most preferred encoding method is one hot encoding. because all are nominal categories and there is no ranking among the values. It provides a more accurate representation of the dataset to the machine learning algorithm.